

# A 500-Basis-Point Treasury Stress

## When Refinancing, Inflation, and Financing Capacity Feed Back

U.S. Fiscal Dynamics Simulator

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### Abstract

Can the United States continue borrowing to finance primary deficits and rising interest expense indefinitely, or does the interaction eventually force a change in regime? This paper asks that question with a family of quarterly scenarios produced by the same Treasury debt engine. The main stress scenarios begin in 2026Q4 with a 500-basis-point premium on new Treasury issuance for ten years and a 200-basis-point private-credit spread for two years. Existing fixed-rate debt retains its coupon until maturity. New interest expense is allocated to domestic private, foreign, and Federal Reserve holders; a configurable share enters demand; inflation affects the Federal Reserve policy rule; and policy rates feed back into subsequent Treasury interest.

An artificial control sets the spending of stress-created interest to zero, isolating the Treasury refinancing arithmetic. Mandatory quarterly clearing then keeps inflation near 2 percent while debt held by the public reaches 252 percent of GDP in 2056. The main case combines official 2026 holder shares with illustrative spending propensities: domestic private recipients spend 25 percent of incremental interest on domestic output and foreign recipients spend 5 percent. Together, those assumptions send 15.67 percent of incremental cash interest into domestic demand. A grid of Federal Reserve responses keeps inflation below a 4-percent early-warning line through 2056 only when the effective spending share stays below 10 percent. At the main-case 15.67-percent share, inflation remains above 4 percent beginning in 2037Q3, passes 6 percent in 2048, and reaches 8.04 percent in 2056 while still rising. A published debt–yield sensitivity accelerates the path: inflation remains above 8 percent beginning in 2051Q3, above 10 percent beginning in 2055Q1, and reaches 11.26 percent in 2056 alongside a 19.92-percent policy rate.

A finite aggregate financing-capacity rule supplies the second test. Under its central sensitivity, required issuance exceeds 1.5 times reference capacity in 2040Q3, or 2039Q3 when debt-dependent yields are active. A tight 1.25-times ceiling binds in 2034Q3; a wide 2.0-times ceiling remains slack through 2056. The 4-percent date is the first warning in an inflation path that continues to accelerate. The financing-capacity date is a hard model stop: automatic private placement has ended and a new fiscal, monetary, regulatory, or payment regime must begin.

The paper then gives that break a conditional recovery. The central private-clearing path ends 2039Q2 with inflation at 4.89 percent, debt at 178.82 percent of GDP, annualized interest expense at 20.49 percent of GDP, and the Fed’s short-rate setting at 8.72 percent. Beginning in 2039Q3, a joint settlement is assumed to lower marginal Treasury rates gradually, permit inflation to peak at 5.2 percent, and cause one year of negative 1-percent real growth. The cohort solver finds that a permanent primary-balance improvement of 5.48 percent of GDP, phased in over four years, returns debt/GDP to 178.82 percent by 2056 and leaves it falling. Debt first peaks at 209.27 percent in 2043; interest expense declines to 8.88 percent of GDP and inflation to 2.5 percent by 2056. The same favorable growth, inflation, and rate paths without the fiscal package end at 265.14 percent debt/GDP. The recovery’s macro and yield paths are declared assumptions; the required fiscal adjustment and Treasury cash flows are the computed results. Business as usual breaks, but a joint settlement reached at the financing boundary prevents the later double-digit-inflation endgame.

# 1 What would make federal borrowing break?

The parable in Peter and Andrew Schiff’s *How an Economy Grows and Why It Crashes* ends with foreigners acquiring claims on domestic wealth [14]. Ray Dalio describes debt cycles that culminate in monetization, inflation, domestic conflict, reserve-currency decline, and geopolitical change [12, 13]. Those stories ask the question studied here: if the federal government repeatedly borrows to cover the primary deficit—the gap between noninterest spending and revenue—and an ever-larger interest bill, what eventually forces the process to change?

**One model, several scenarios.** This paper investigates that question with a quarterly computer model of U.S. Treasury debt. The model begins with a reconstructed 2026 stock of bills, notes, bonds, TIPS, and floating-rate notes. In every quarter it calculates coupon and principal payments, refinances maturing securities, and issues additional debt for the primary deficit and cash interest. It then connects that Treasury ledger to a simple economic loop: some interest payments become spending; spending affects output and inflation; the Federal Reserve raises rates in response; and those rates increase the cost of the next round of Treasury borrowing.

The paper runs several scenarios through that common model. The main stress scenarios add five percentage points to new Treasury yields from 2026Q4 through 2036Q3 and add two percentage points to private-credit rates for the first two years. A no-shock comparison, an artificial zero-spending control, alternative bondholder-spending shares, a grid of Federal Reserve reactions, debt-dependent Treasury yields, and three buyer-capacity curves isolate different parts of the mechanism. Each reported date belongs to one of these stated conditional scenarios.

**What can stop automatic rollover?** The Treasury ledger by itself can run forever. If every auction is guaranteed to clear and the government may issue whatever amount balances its accounts, each quarter’s financing need simply becomes debt carried into the next quarter. Debt and interest can reach extraordinary levels while the program continues producing another number. A meaningful breaking point therefore requires an economic limit outside the accounting identity.

The first diagnostic tracks inflation control under the initiating stress. Inflation moves freely while a Federal Reserve reaction rule raises rates in an attempt to return it toward the reference path near 2 percent. For reporting, the paper marks the start of the first run lasting at least four consecutive quarters above 4 percent as an early warning that the rule is failing to pull inflation back. The economic severity comes from the subsequent path: whether inflation retreats or continues through 6, 8, and 10 percent while policy rates rise with it.

The second diagnostic imposes a hard financing limit. Gross issuance in a quarter must cover maturing principal as well as the primary deficit and interest. The financing-capacity rule begins with the amount buyers absorb along the reference path. Higher yields attract additional buyers until a specified maximum multiple of that reference amount is reached. In the central curve, each 100-basis-point yield increase adds capacity equal to 5 percent of reference issuance, up to a total capacity of 1.5 times the reference amount. When required issuance exceeds that maximum, the simulation stops: the next quarter requires Federal Reserve purchases, fiscal adjustment, compulsory absorption, inflationary accommodation, or a change to promised payments.

**What the scenarios show.** The central macro result is a progressive loss of inflation control. An artificial control that prevents stress-created interest from entering demand produces much more debt but little inflation. In the main case, domestic private recipients spend 25 percent of incremental interest

on domestic output and foreign recipients spend 5 percent. Weighting those rates by official holder shares sends 15.67 percent of incremental cash interest into domestic demand. That share lies above the model’s 9.5-to-10-percent early-warning threshold. Inflation remains above 4 percent beginning in 2037Q3, above 6 percent beginning in 2048Q1, and reaches 8.04 percent in 2056. Raising rates eventually adds enough Treasury interest to reinforce the demand pressure, and every Federal Reserve rule tested at the main spending share ends with inflation pressure still rising. Adding debt-dependent yields produces sustained inflation above 8 percent in 2051Q3 and above 10 percent in 2055Q1.

The separate financing-capacity experiment supplies the hard break. Its central curve stops the run in 2039Q3 when debt-dependent yields are active; tighter and wider curves move that date from 2034Q3 to beyond 2056. The model therefore identifies the mechanisms that can end automatic debt rollover, while the dates remain conditional on spending and buyer-capacity parameters that still require empirical calibration. CBO similarly emphasizes that the timing of a U.S. fiscal crisis depends on more than debt/GDP alone [3].

The recovery experiment asks what happens if Washington acts at that break instead of continuing the unchanged-policy path. It gives the economy a short recession, holds inflation between 2.5 and 5.2 percent, and lets newly issued Treasury rates decline over several years as the settlement gains credibility. Under those stated conditions, the cohort engine requires a permanent 5.48-percent-of-GDP primary-balance improvement phased in over four years. Debt rises for another four years because high-coupon securities remain in the stock, peaks at 209 percent of GDP, and then declines to 179 percent in 2056. The result still leaves an unprecedented debt burden and a transformed federal budget. It also shows the size and duration of the adjustment that prevents the crisis from becoming the country’s permanent economic regime.

## 2 Starting point and initiating stress

The experiment starts in **2026Q4**. Opening debt held by the public is approximately \$32.2 trillion, or 99.2 percent of GDP. The cohort ledger tracks bills, fixed-rate notes and bonds, Treasury Inflation-Protected Securities (TIPS), and floating-rate notes (FRNs) by principal, coupon, and maturity. It also carries a coarse nonmarketable public-debt category. The fiscal and economic reference paths come from CBO’s February 2026 ten-year baseline and long-term extension [1, 2].

The initial stress adds five percentage points to each new-issuance rate for forty quarters. In 2026Q4, the bill rate therefore rises from 3.278 to 8.278 percent, the note and bond rate from 4.197 to 9.197 percent, and the TIPS real rate from 1.750 to 6.750 percent. FRNs reset from the higher bill reference rate plus their fixed spreads. Existing notes and bonds keep their contractual coupons until maturity. The premium disappears after 2036Q3; subsequent rate movements come from the policy and debt-yield rules.

A 200-basis-point private-credit spread applies for the first eight quarters and enters the output equation directly as a temporary recessionary impulse. A structural Treasury-to-mortgage pass-through lies outside the experiment. The 500-basis-point Treasury premium is also a scenario trigger. It represents an outer-tail repricing large enough to test the system. Its probability is unspecified, and the experiment treats it as a sensitivity rather than a 2026 auction forecast.

| Element                   | Treatment                                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Horizon                   | 2026Q4 through 2056Q3; no terminal debt target.                                                                                |
| Treasury stress           | 500 basis points on new issuance for forty quarters.                                                                           |
| Private-credit stress     | 200 basis points for eight quarters.                                                                                           |
| Inherited fixed-rate debt | Contractual coupons continue until maturity.                                                                                   |
| Primary deficits          | CBO reference path, scaled with scenario nominal GDP and adjusted by an aggregate automatic stabilizer.                        |
| Inflation                 | Evolves freely from the reference path, persistence, and the lagged output gap; the model records the full path without a cap. |
| Federal Reserve           | A smoothed price-stability reaction changes the policy-rate proxy and passes through to new Treasury rates.                    |
| Treasury financing        | Cash interest, primary deficits, and maturing principal generate new issuance each quarter.                                    |

These rules govern the unresolved stress path. The recovery experiment changes course in 2039Q3. It publishes a post-settlement path for growth, inflation, and marginal Treasury rates, then solves for the primary-balance improvement required to return debt/GDP to its 2039Q2 level by 2056 and leave the ratio falling. That terminal condition is explicit; it belongs only to the recovery solver, while the unresolved stress scenarios remain untargeted.

The initiating rate shock reaches federal cash expense gradually. Bills roll quickly and FRNs reset; fixed-rate securities reprice only as they mature. Each quarter the model indexes TIPS, accrues interest, matures cohorts, refinances principal, and issues additional debt for the primary deficit and cash interest. Securities issued at quarter-end begin adding interest in the next quarter. This cohort structure is the accounting kernel shared by every experiment below.

### 3 How the stress unfolds

Markets never label a move an “exogenous shock.” Investors, politicians, and television anchors attach a story to it, even when they disagree about the underlying cause. The datelines and news bulletins below place one plausible public history around a crisis and a recovery. Through 2039Q2, bondholders spend 15.67 percent of their extra interest income on domestic output, a widening debt/GDP gap raises longer-term yields, and the central buyer curve requires progressively larger auction spreads. The curve reaches its limit in 2039Q3. Washington then changes course: the Fed bridges the market, Congress enacts a four-year fiscal package, and marginal Treasury rates decline as the package gains credibility.

The recovery is conditional in a precise way. Its quarterly paths for inflation, real growth, and new Treasury rates are published assumptions. The cohort engine determines how the inherited securities refinance, how much interest they generate, and how debt evolves. It also solves for the permanent primary-balance improvement required to return debt/GDP to its pre-settlement level by 2056 with the ratio falling. The people, legislation, market reactions, and quotations in the chronology are invented. The numerical statements before the settlement come from the sequential crisis run; those after it come from the conditional recovery run.

In this history, the panic begins when budget negotiations collapse just before Treasury exhausts its borrowing authority. Moody’s, which lowered the United States from Aaa to Aa1 in May 2025, announces a second one-notch cut, from Aa1 to Aa2 [4]. A poorly received Treasury auction then leaves dealers holding far more securities than planned. The stress test fixes the combined repricing at five percentage

points. The downgrade, legal threat, and auction seizure supply the public explanation for an outer-tail move of that imposed size: investors suddenly demand compensation for payment delay, inflation, and the danger that a seized-up market will leave them unable to resell Treasuries.

### **Wednesday, October 14, 2026—the break.**

*Breaking News, 4:18 p.m.: A Treasury auction has cleared only after the largest one-day yield increase in modern market history. The selloff follows the collapse of congressional budget talks and Moody's downgrade of the United States from Aa1 to Aa2. The White House says every federal payment will be honored. Banks are repricing mortgages and corporate credit tonight.*

Within days, buyers demand five additional percentage points to hold newly issued Treasury debt. The bill rate jumps from 3.278 to 8.278 percent; note and bond rates jump from 4.197 to 9.197 percent. Private-credit rates rise by two percentage points as banks and bond investors reprice newly quoted dollar loans. Stocks sell off because Treasury yields form the base discount rate used to value future corporate earnings: adding five percentage points lowers the present value of those earnings, while dearer private credit points toward less investment and lower profits. Existing bonds also fall in market value, imposing immediate losses on banks, funds, and leveraged holders. Lenders pull mortgage rate sheets while they reprice, and nightly news panels debate whether the United States is insolvent or merely politically incapable of governing itself. These asset-price reactions belong to the invented news history; the macro equation receives only the stated two-point private-credit spread.

The federal interest bill moves slowly. Old notes and bonds still carry their contractual coupons, so bills, floating-rate notes, and the first newly issued securities take the blow. Inflation ends December at 2.37 percent and debt held by the public at 99.65 percent of GDP. To most households, this still looks like a violent financial-market event rather than a federal budget emergency.

### **Thursday, December 12, 2030—the headlines declare victory too soon.**

*Evening News: Inflation is back near 2 percent and private borrowing has begun to recover. Administration officials say the economy has absorbed the Treasury shock. A quieter number in today's budget report shows federal interest expense above 8 percent of GDP.*

Bills have rolled repeatedly, and progressively larger cohorts of notes and bonds have matured. Treasury replaces them at crisis rates while also borrowing to cover the primary deficit and the larger interest bill. Debt reaches 116.28 percent of GDP. During this quarter, the higher-rate debt generates \$420 billion more cash interest than the economy would have seen without the panic. Bondholders spend \$65.9 billion of that additional income on domestic output. Interest expense reaches 8.36 percent of annualized GDP while inflation is still only 2.21 percent. The widening debt gap has also added 33 basis points to the long-bond rate. The apparent calm conceals a much more expensive debt stock.

### **Thursday, September 18, 2036—the feedback becomes visible.**

*Market Close: Treasury bill rates have reached 11.75 percent after the Federal Reserve's latest increase. Inflation is 3.77 percent. Lawmakers are demanding lower rates while Fed officials warn that cutting now would reignite price growth.*

The private-credit recession has faded, but the federal interest transfer is now supporting demand. The Fed's short-rate setting has reached 6.75 percent. The original five-point crisis premium still sits on top,

producing an 11.75-percent bill rate. The debt–yield response adds another 1.37 percentage points to the long-bond rate, which reaches 12.57 percent. Each Fed increase depresses homebuilding, business investment, and consumer credit; it also raises the coupon Treasury promises when the next security is sold.

Debt has reached 154.83 percent of GDP. Interest expense absorbs 15.99 percent of annualized GDP and the total deficit is 17.06 percent of GDP. During this quarter alone, stress-created interest sends another \$222 billion into domestic demand. The central bank is pressing the brake while the Treasury interest bill is pressing the accelerator.

### **Wednesday, October 1, 2036—Washington removes the panic premium.**

*Breaking News: Congress has replaced recurring debt-limit votes with permanent borrowing authority for obligations already enacted. The Federal Reserve has announced a standing Treasury-market facility intended to prevent another auction seizure. Treasury yields fall, but dealers warn that the volume of securities still coming to market is overwhelming traditional buyers.*

This is why the extra five percentage points disappear in this history. Congress removes the recurring legal threat to payment, and the Fed reduces the risk that dealers will be left without market liquidity. The five-point premium vanishes, but a 3.70-point auction- clearing spread takes its place as Treasury asks private buyers to absorb ever more paper. The bill rate therefore falls only from 11.75 to 10.62 percent. Washington has repaired the original panic while leaving the primary deficit and ten years of high-coupon debt in place. Inflation risk and financing pressure survive the rescue.

### **Thursday, September 17, 2037—the Fed raises rates again.**

*Fed Raises Benchmark Again: With inflation moving above 4 percent, the central bank has lifted its short-rate setting to 7.41 percent. Officials say high real borrowing costs will cool demand, but acknowledge that rising Treasury interest payments are weakening the effect.*

The Fed has continued tightening after the panic premium vanished: its short-rate setting reaches 7.41 percent one year later. Its reaction rule responds to inflation with gradual quarterly adjustments rather than a single enormous move. Officials expect the usual channel—dearer mortgages, auto loans, and corporate credit—to cool spending without causing a depression.

That channel is working, but another one is pushing the other way. Treasury must refinance at the Fed's higher rates, and bondholders spend part of the resulting interest income. Debt has reached 163.40 percent of GDP and interest expense is 17.10 percent of GDP. The bill rate offered to buyers is 11.33 percent after the market-clearing spread. September begins a run of at least four consecutive quarters above 4 percent inflation. The Fed keeps raising rates in this history. Its dilemma is already visible: every increase removes some private demand now and adds to federal interest income as Treasury refinances later.

### **Thursday, June 16, 2039—the market reaches the edge.**

*Treasury Buyers Demand 13-Percent Bill Rates: The government expects to sell \$26.6 trillion of securities this quarter, including repeated refinancing of short-term debt. Dealers say the autumn financing calendar could exceed the amount private portfolios can absorb at any yield contemplated by current policy.*

Inflation is 4.89 percent, the Fed's short-rate setting is 8.72 percent, and the additional auction-clearing spread has reached 4.63 percentage points. New bills yield 13.35 percent. Debt stands at 178.82 percent of GDP, interest expense at 20.49 percent of annualized GDP, and the total deficit at 21.20 percent. Required issuance is already 1.488 times the reference amount under the paper's central buyer curve.

That curve fails in the next quarter. It can absorb \$27.303 trillion, while Treasury needs to place \$27.341 trillion, leaving \$37.4 billion without a private buyer. The dollar shortfall is small beside the auction, but its meaning is large: another increase in yield has reached the curve's assumed capacity ceiling. Borrowing the missing amount would add to the securities requiring buyers. A temporary Federal Reserve facility can bridge the auction; permanent financing now requires lower primary borrowing, a larger captive buyer base, monetary accommodation, or changed payments.

#### **Thursday, September 15, 2039—Congress chooses the settlement.**

*President Signs Four-Year Fiscal Accord; Fed Opens Temporary Treasury Facility: The law combines a broad consumption levy, additional income and payroll revenue, slower growth in retirement and health benefits, and limits on other spending. Leaders of both parties say the package is intended to end emergency central-bank support by restoring private demand for federal debt.*

At full implementation, the law improves the primary balance by 5.48 percent of GDP relative to the previous path. The change arrives in sixteen equal quarterly steps, spreading a fiscal adjustment with few peacetime U.S. precedents over four years. The fictional law protects low-income transfers and public investment while distributing the remainder broadly; the simulator represents only its aggregate primary-balance effect. Legislative scoring of the invented components would require a separate tax-and-benefit model.

The Fed provides market liquidity while the first measures take effect, and the announced fiscal path brings private buyers back. New bill, note, and long-bond rates fall to 8.50, 9.00, and 9.00 percent. Those rates are published recovery assumptions, followed by a gradual decline rather than an overnight return to normal. The first-quarter fiscal improvement is only 0.34 percent of GDP. Interest expense reaches 21.01 percent of GDP, the total deficit 22.36 percent, and debt 182.74 percent. The rescue has stopped the auction seizure; years of high-coupon debt remain to be worked through.

#### **Thursday, September 20, 2040—the price of stabilization.**

*Inflation Peaks at 5.2 Percent During Second Year of Contraction: Unemployment is rising, households are absorbing the new tax system, and Congress has rejected proposals to suspend the fiscal accord. Treasury bill rates have declined to 7.5 percent.*

Real GDP is contracting at a 1-percent annual rate and inflation reaches the recovery path's 5.2-percent peak. The fiscal improvement has reached 1.71 percent of GDP, enough to bring the primary budget approximately into balance. Interest from the inherited stock still equals 19.67 percent of GDP, leaving a 19.61-percent total deficit and debt at 195.42 percent of GDP. The public sees a recession, higher taxes, slower benefit growth, and continuing inflation at the same time. The political bargain survives because the Fed's bridge is temporary, payments continue, and each quarter's new debt carries a lower rate than the stock it is beginning to replace.

#### **Thursday, June 18, 2043—debt peaks after the crisis has ended.**

*Debt Reaches 209 Percent of GDP, Yet Treasury Yields Fall Again: Budget officials report a 3.9-percent primary surplus and say the debt ratio is at its turning point. Inflation has declined to 4.3 percent and real growth has recovered above 2 percent.*

The fiscal package is now fully phased in. New bills yield 5.24 percent and long bonds 6.04 percent, while the inherited marketable stock still carries an average effective rate of 7.81 percent. Interest expense is 17.24 percent of GDP. The government therefore runs a 13.37-percent total deficit even with the large primary surplus. Debt/GDP nevertheless peaks at 209.27 percent: nominal output is growing fast enough, and new borrowing slowly enough, for the denominator to begin overtaking the debt stock. Recovery means reversing the ratio's direction long before the nominal debt is repaid.

#### **Thursday, December 19, 2047—the emergency becomes a durable compact.**

*Debt Ratio Falls for a Fourth Year as Inflation Recedes to 3.3 Percent: Congress renews the core budget rules after an election fought over how to divide the continuing primary surplus. The Federal Reserve reports that emergency Treasury support has ended.*

Debt has declined to 202.08 percent of GDP. Interest expense remains enormous at 12.03 percent of GDP, but the 3.95-percent primary surplus reduces the total deficit to 8.08 percent. Real growth is 1.94 percent, inflation is 3.34 percent, bills yield 4.56 percent, and long bonds 5.36 percent. The consumption levy and benefit changes have become ordinary, unpopular features of federal policy. Markets treat them as durable enough to finance the remaining deficit without the emergency facility.

#### **Thursday, September 14, 2056—America has recovered, with scars.**

*Inflation Near 2.5 Percent as Debt Falls to Its 2039 Share of the Economy: Treasury reports another year of declining debt/GDP and no payment disruption since the crisis. The federal interest bill remains close to 9 percent of GDP, preserving intense pressure on every budget.*

Debt ends at 178.82 percent of GDP and has fallen 2.55 percentage points during the preceding four quarters. Inflation is 2.5 percent, real growth 1.7 percent, new bills yield 4.3 percent, and long bonds 5.1 percent. The average effective rate on the marketable stock has declined to 4.68 percent as the old crisis cohorts mature. Interest expense is still 8.88 percent of GDP. A 4.03-percent primary surplus leaves a 4.85-percent total deficit that nominal growth can absorb while the ratio continues downward.

Prices have risen 82 percent over the seventeen-year recovery window. How nominal wages, asset values, taxes, and benefits divide that burden lies beyond these equations. The relatively happy ending is narrow and concrete: auctions function, inflation returns near its old range, real output grows, no Treasury payment is missed, and the debt ratio has been turned. America survives by replacing its former fiscal promises with a permanent compact large enough to support the debt it accumulated.

## **4 Where the crisis and recovery come from, in graphs**

The first three figures show the unresolved stress path. They explain the pressure that forces the 2039 settlement and show where the system goes when policy never changes. The fourth figure begins at the regime switch and displays the conditional recovery. Separating the two pieces preserves an essential distinction: the pre-settlement feedback is generated inside the sequential model, while post-settlement growth, inflation, and marginal yields are declared paths used to calculate the required fiscal closure and Treasury cash flows.

## The imposed shock gives way to self-sustaining rate pressure

The left panel of Figure 1 contains two scenario instructions and one calculated result. The gold line supplies the 500-basis-point Treasury premium for ten years. The gray line supplies the 200-basis-point private-credit spread for two years. The red line is produced inside the run: as debt/GDP rises above the no-shock path, the central debt–yield rule adds an increasing premium to long bonds. By the time the original panic premium ends in 2036, the debt premium is already 137 basis points; it reaches 464 basis points by 2056.

The right panel shows the rates paid on new securities. Bills take the entire five-point shock and the full change in the Fed’s short-rate setting. Long bonds begin from a higher reference rate, receive the panic premium through 2036, and subsequently retain pressure from the widening debt ratio. The sudden drop in 2036 is therefore visible without ending the long-run rise.

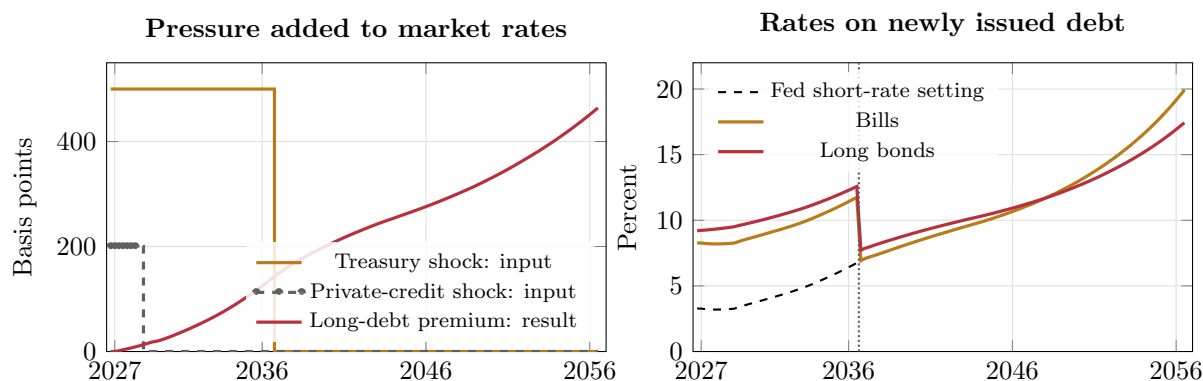


Figure 1: The rate sequence in the unresolved stress path. The two rectangular paths in the left panel are imposed stresses. The rising red path and every rate in the right panel are calculated before any recovery settlement. The vertical dotted line marks the removal of the five-point Treasury premium after 2036Q3.

## Refinancing turns high new rates into a federal budget problem

Figure 2 shows the slow-moving Treasury mechanism. Only 27 percent of marketable debt has repriced by the end of the first shock quarter because old notes and bonds retain their coupons. The repriced share exceeds 90 percent in 2035 and reaches 100 percent in 2055. This delay explains why the interest bill keeps climbing long after the initial market break.

Debt rises from approximately 100 percent to 290 percent of GDP. Interest expense and the total deficit converge near 46 percent of annualized GDP. Interest expense slightly exceeds the total deficit at the endpoint because the primary balance has moved into a small surplus; the remaining deficit is entirely interest-driven. Gross issuance rises even faster in dollar terms. It includes every bill or other security sold to replace maturing principal, so the same short-term borrowing can appear several times during a year; the flow reaches \$194.9 trillion in 2056Q3, while the debt stock increases only by the net financing need.

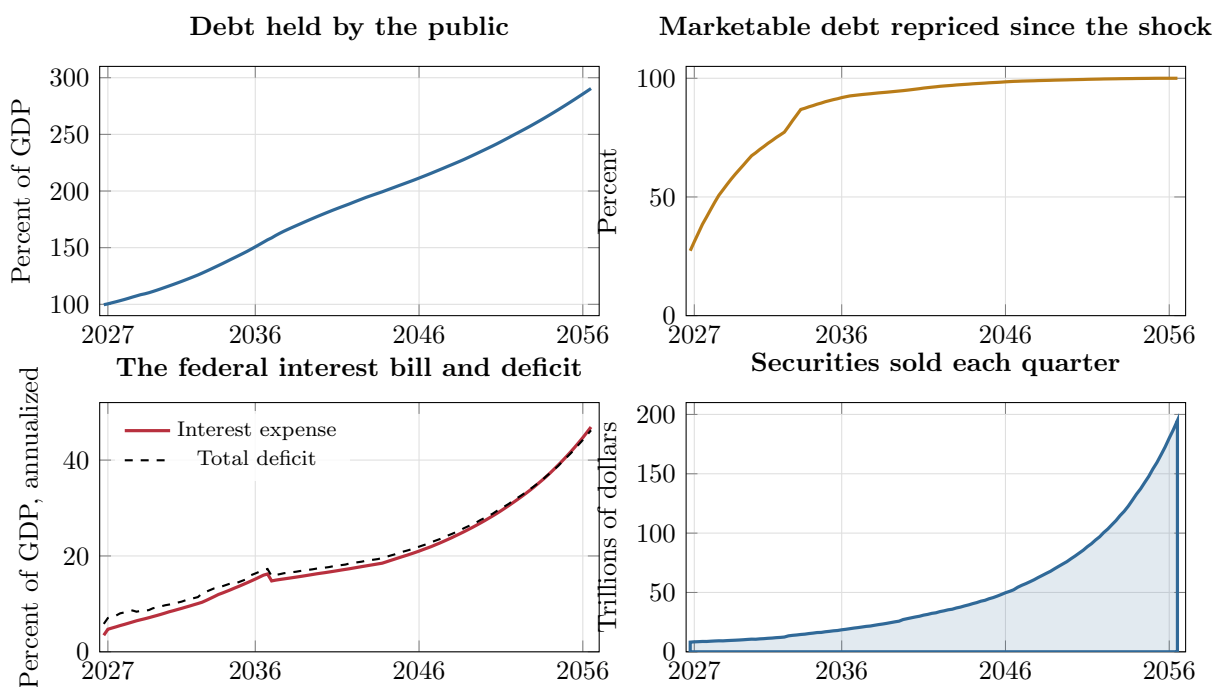


Figure 2: How refinancing transmits the rate stress into debt, interest expense, deficits, and auction volume. Gross issuance is a flow that includes repeated replacement of maturing principal; debt held by the public is a stock.

## Interest payments weaken the Fed’s brake

The last set of panels closes the loop. The first plots the portion of stress-created interest that domestic and foreign recipients spend on U.S. output each quarter. That amount is 15.67 percent of the GDP-comparable incremental cash interest calculated by the parallel no-shock ledger. It grows from nearly zero in 2026 to \$3.91 trillion in 2056Q3.

The demand impulse initially only offsets part of the private-credit contraction. It later pushes output above the reference path, as the middle panel shows. Inflation consequently separates from the reference path in the right panel. The Fed continues raising rates, but those rate increases feed the Treasury interest transfer shown here. Inflation crosses 4 percent in 2037, 6 percent in 2045, 8 percent in 2051, and 10 percent in 2055.

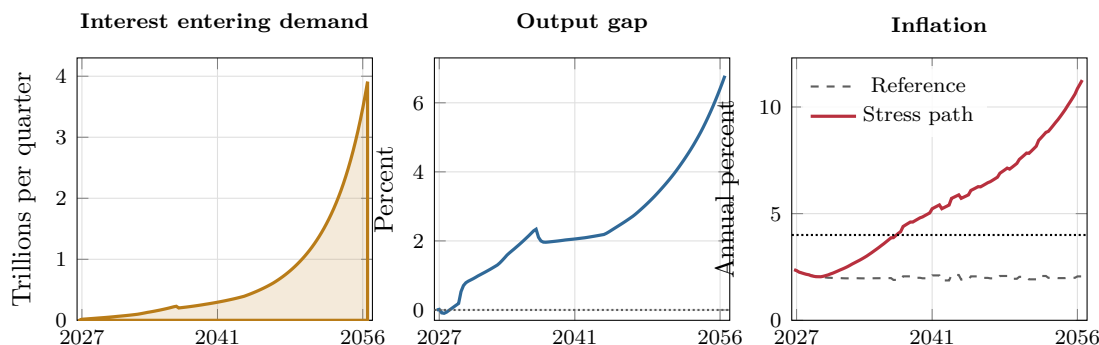


Figure 3: The fiscal-monetary feedback in pictures. Higher rates create additional cash interest; the assumed spendable share supports demand; output and inflation move above their reference paths; and the Fed responds with rates that make the next refinancing round more expensive. The dotted line in the last panel is the paper’s 4-percent early-warning level.

## The settlement bends the path slowly

Figure 4 begins after the private-capacity warning. The left panel shows why the rescue never feels miraculous: debt keeps rising for almost four years as the high-rate stock rolls forward, then turns down. The dashed line applies the same assumed growth, inflation, and marginal-rate recovery without the fiscal package. It reaches 265 percent of GDP in 2056. Restored confidence alone leaves the debt ratio climbing.

The middle panel shows the two flows that make the turn possible. The primary balance moves from deficit to a surplus near 4 percent of GDP while interest expense falls from 21 to 9 percent as crisis cohorts mature. The right panel displays the declared inflation and rate paths. Their gradual normalization is an input to the recovery calculation, while the debt, interest, and required 5.48-percent fiscal adjustment are outputs of the cohort engine and closure solver.

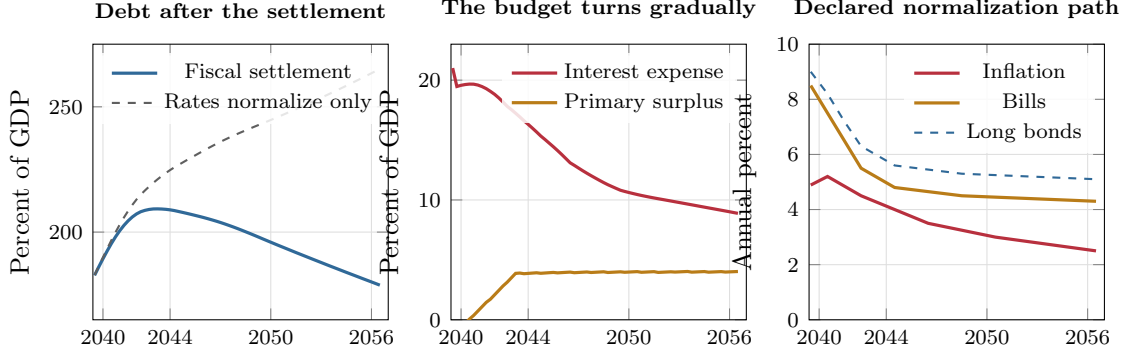


Figure 4: The conditional recovery after the 2039 financing boundary. The left panel compares the solved fiscal settlement with the same macro and rate assumptions absent the primary-balance improvement. Inflation and marginal rates in the right panel are scenario assumptions; the cohort engine calculates the two debt paths and the interest path.

## 5 The sequential fiscal–monetary loop

The sequential experiment preserves a parallel no-shock Treasury ledger. Only cash interest created by the stress enters the new demand channel. Comparing interest as a share of scenario and reference GDP prevents a higher scenario price level from being mistaken for additional real purchasing power.

The starting allocation of interest recipients uses approximate 2026 official aggregates. CBO reports \$32.095 trillion of FY2026 debt held by the public and \$4.540 trillion held by the Federal Reserve. Treasury’s June 2026 TIC table reports \$9.299 trillion of foreign holdings [6]. The residual allocation is 56.88 percent domestic private, 28.97 percent foreign, and 14.15 percent Federal Reserve. The timing and valuation conventions differ, so these aggregates serve as benchmark shares. A security-level ownership reconciliation lies outside this calibration.

In the central case, domestic private recipients spend 25 percent of incremental cash interest on domestic output, foreign recipients spend 5 percent domestically, and the Federal Reserve share contributes nothing directly to demand. The effective fraction is therefore

$$m_I = (0.5688)(0.25) + (0.2897)(0.05) + (0.1415)(0) = 0.1567.$$

The domestic and foreign spending fractions remain scenario assumptions. Research on the redistribution and interest-rate-exposure channels supports modeling heterogeneous marginal propensities to consume. This experiment supplies the aggregate coefficient [10]. The BIS likewise notes that higher government interest payments to private bondholders can support demand and weaken monetary transmission when public debt is high [11].

Let  $x_t$  be the output gap,  $a_t$  the policy-rate adjustment relative to the reference path,  $d_t^P$  the automatic-stabilizer primary-deficit impulse as a share of GDP, and  $d_t^I$  the spendable incremental-interest impulse. The central quarterly rule is

$$x_{t+1} = 0.80x_t - 0.25\Delta r_{t-1}^{real} - 0.15s_{t-1}^{private} + 0.80d_t^P + 1.00d_t^I.$$

Inflation is represented as a deviation from the CBO reference path:

$$\pi_{t+1} = \pi_{t+1}^0 + 0.98(\pi_t - \pi_t^0) + 0.06x_{t-1}.$$

The policy adjustment follows

$$a_t = 0.50a_{t-1} + 0.50 \left[ 1.50(\pi_t - \pi_t^0) + 0.50x_t \right].$$

The short Treasury rate receives the full policy adjustment; notes receive 75 percent and long bonds and TIPS real rates receive 50 percent. An output gap of  $x_t$  changes the quarterly primary deficit by  $-0.45x_t$  times annualized potential nominal GDP divided by four. These equations make the mechanism reproducible. Their coefficients are reduced-form sensitivities; a fully estimated U.S. macroeconomic model remains future work.

## 6 Does monetary tightening stop the loop?

Table 1 separates the accounting channel from the interest-income demand channel. The 500-basis-point premium lasts ten years in every stress row. The central and high-spending rows also include the temporary private-credit spread.

Table 1: Sequential stress through 2056Q3

| Scenario                      | Spendable<br>interest share | First sustained<br>> 4% warning | 2056<br>inflation | Peak policy<br>rate | 2056<br>debt/GDP |
|-------------------------------|-----------------------------|---------------------------------|-------------------|---------------------|------------------|
| No shock                      | —                           | none                            | 2.06%             | 3.28%               | 172.77%          |
| Stress, zero-spending control | 0.00%                       | none                            | 2.05%             | 3.28%               | 251.56%          |
| Stress, central spending      | 15.67%                      | 2037Q3                          | 8.04%             | 13.66%              | 255.61%          |
| Stress, high spending         | 31.34%                      | 2033Q4                          | 40.93%            | 78.21%              | 235.59%          |

The zero-spending control isolates the accounting channel from macroeconomic transmission. It sets every recipient’s spending propensity to zero solely to measure what refinancing arithmetic can generate on its own. Higher rates raise 2056 debt/GDP by almost 79 percentage points relative to the no-shock path, and peak interest expense reaches 13.43 percent of GDP. Inflation stays near the reference path while the mandatory-clearing engine continues placing the government’s debt.

The central spending row activates that missing route. Interest income supports output; inflation rises; the Federal Reserve raises rates; new Treasury financing becomes more expensive; and the next interest transfer becomes larger. By 2056, interest expense reaches 28.72 percent of GDP and the total deficit reaches 29.40 percent. Inflation pressure and debt/GDP are both still rising at the endpoint.

The 2037Q3 date marks the first sustained warning. The inflation problem then deepens. In the central path, inflation remains above 6 percent beginning in 2048Q1 and reaches 8.04 percent in 2056. With the central debt–yield feedback, it remains above 8 percent beginning in 2051Q3, above 10 percent beginning in 2055Q1, and reaches 11.26 percent at the endpoint.

The high-spending row is a nonlinear stress test. Inflation enlarges nominal GDP enough to produce a lower 2056 debt ratio even as interest exceeds 112 percent of GDP. The most extreme full-domestic-spending case reaches the model’s 99-percent rate boundary and is excluded from the main table.

### The 4-percent early-warning frontier

The policy frontier varies the effective share of incremental interest entering demand and the Federal Reserve inflation-response coefficient. For each spending share, the selected row minimizes the peak policy rate among paths that stay below the 4-percent warning line through 2056. If every path crosses the line, the table reports the lowest peak inflation.

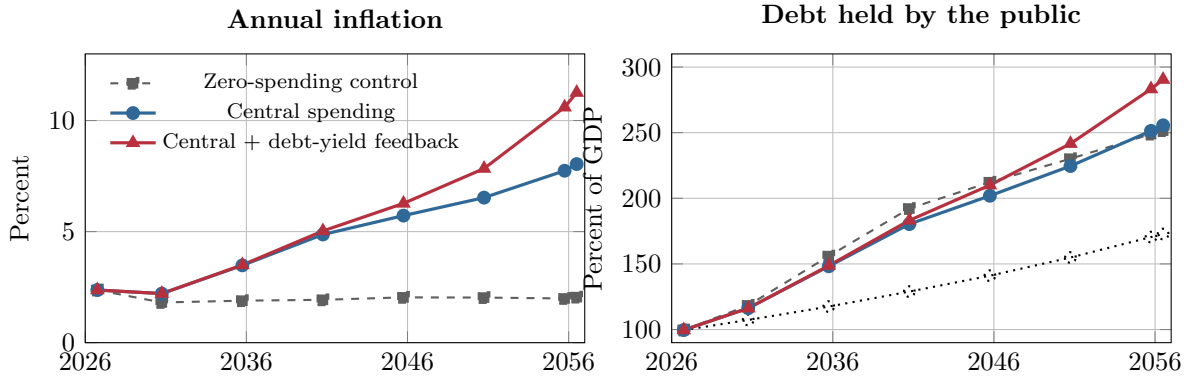


Figure 5: Sequential paths. The left panel shows the interest-income demand channel and the additional debt-yield feedback. The right panel shows that a lower debt ratio in an inflationary case can coexist with a more severe interest and price-stability problem.

| Interest to demand | Selected response | Below 4% through 2056 | Peak inflation | Peak policy rate |
|--------------------|-------------------|-----------------------|----------------|------------------|
| 5.0%               | 0.5               | yes                   | 3.42%          | 4.04%            |
| 8.0%               | 1.0               | yes                   | 3.90%          | 5.29%            |
| 9.0%               | 1.5               | yes                   | 3.99%          | 6.40%            |
| 9.5%               | 3.0               | yes                   | 3.99%          | 9.39%            |
| 10.0%              | 2.5               | no                    | 4.29%          | 9.26%            |
| 15.67%             | 0.5               | no                    | 6.96%          | 6.41%            |

The early-warning boundary in this grid lies between 9.5 and 10 percent. Stronger tightening becomes counterproductive after the interest-transfer channel grows large enough: the policy rate suppresses private demand while raising the government’s refinancing cost and the income paid to bondholders. Even the positive-spending paths that stay below 4 percent end with both inflation pressure and debt/GDP rising. Their endpoint trends show that the Federal Reserve has postponed the warning without producing convergence by 2056.

## 7 Debt deterioration can sustain the rate pressure

The initiating premium disappears after ten years. A second experiment lets the debt path itself affect later Treasury yields. Bhatt, Diercks, Eyal, and Skaperdas estimate that a one-percentage-point increase in expected U.S. debt/GDP raises the longer-run neutral rate by about 1–2 basis points and the ten-year term premium by about 2–3 basis points [7]. The model applies lower, central, and upper translations to the lagged scenario debt/GDP gap relative to the parallel reference path.

The central translation adds 1.5 basis points per debt-ratio point to note and TIPS real rates and 4 basis points per point to long-bond rates; bills receive no debt adjustment. The source estimates expected debt effects. The published estimate anchors the scale; this experiment supplies a one-quarter lag, a realized debt gap, and the maturity mapping.

| Debt-yield sensitivity | 2056 inflation | Peak policy rate | 2056 debt/GDP | Long-yield addition |
|------------------------|----------------|------------------|---------------|---------------------|
| None                   | 8.04%          | 13.66%           | 255.61%       | 0 bp                |
| Lower                  | 10.12%         | 17.67%           | 278.54%       | 313 bp              |
| Central                | 11.26%         | 19.92%           | 290.47%       | 464 bp              |
| Upper                  | 12.59%         | 22.57%           | 303.80%       | 645 bp              |

The feedback supplies an ongoing mechanism behind the one-time confidence label. The initial shock creates a debt-ratio gap. That gap raises subsequent financing rates within the published range. Higher interest enlarges debt and spendable bondholder income, producing a larger gap in the next round. A nonlinear jump associated with panic, currency loss, or an auction disruption lies outside the local empirical estimate and the current model.

## 8 A finite financing-capacity boundary

Mandatory placement keeps the path alive even at high yields. The final experiment gives aggregate Treasury financing a finite capacity. Let  $Q_t$  be required gross issuance and  $Q_t^0$  reference issuance scaled to scenario GDP. Required capacity is  $M_t = Q_t/Q_t^0$ . At an aggregate nominal yield gap  $\Delta y_t$ , supplied capacity is

$$K_t = \min\left(\bar{K}, \max\left[0, 1 + \eta \frac{\Delta y_t}{100 \text{ basis points}}\right]\right).$$

The model iterates a common spread across issuance types until  $K_t \geq M_t$ . When  $M_t > \bar{K}$ , no supported yield can place that quarter’s issuance and the run stops.

Three transparent sensitivities are reported. The tight curve adds 2.5 percent of reference capacity per 100 basis points and stops at 1.25 times reference issuance. The central curve adds 5 percent and stops at 1.5 times. The wide curve adds 10 percent and stops at 2 times. These curves are illustrative sensitivities awaiting estimation from Treasury auctions. The reference normalization includes ordinary institutional absorption implicit in the baseline; incremental Federal Reserve intervention is absent.

Table 2: Conditional aggregate financing boundaries

| Capacity                  | Debt-yield feedback | First capacity failure | Result at boundary or 2056                   |
|---------------------------|---------------------|------------------------|----------------------------------------------|
| Tight, $\bar{K} = 1.25$   | no                  | 2034Q3                 | \$5.1B beyond capacity                       |
| Central, $\bar{K} = 1.50$ | no                  | 2040Q3                 | \$145.4B beyond capacity                     |
| Wide, $\bar{K} = 2.00$    | no                  | none                   | 65 bp peak clearing spread; 8.41% inflation  |
| Tight, $\bar{K} = 1.25$   | yes                 | 2034Q3                 | \$11.1B beyond capacity                      |
| Central, $\bar{K} = 1.50$ | yes                 | 2039Q3                 | \$37.4B beyond capacity                      |
| Wide, $\bar{K} = 2.00$    | yes                 | none                   | 10 bp peak clearing spread; 11.27% inflation |

The central 2039Q3 result is the closest this model comes to the requested breaking point. At that date the debt-yield-feedback path requires \$27.341 trillion of gross quarterly issuance, including the refinancing of maturing principal, while the supplied central curve can absorb \$27.303 trillion. The discrete quarter crosses the hard capacity by \$37.4 billion. Debt-dependent yields reduce the separate clearing spread because they have already raised bond rates; the resulting interest cost nevertheless causes the hard capacity to be reached one year earlier.

The next-quarter path at the boundary depends on a new regime: a central bank purchase, fiscal package, regulatory mandate, capital inflow, inflationary accommodation, or delayed payment. Stopping the run

makes that missing choice visible. Continuing to issue debt after the boundary would silently restore the unlimited-absorption assumption that the experiment was designed to test.

## 9 What a Federal Reserve backstop means

Competitive Treasury bidders set the auction's stop-out yield. Federal Reserve rules exclude direct purchases of newly issued securities from Treasury and competitive auction bids [8]. SOMA can roll over maturing holdings through noncompetitive tenders. Those tenders are add-ons to the announced offering and receive the stop-out yield established by competitive bids [9].

The Federal Reserve can support market functioning through repos and secondary-market purchases, as it did during the Treasury-market disruption of March 2020. Such intervention can make an outright failed auction remote. The auction yield and consolidated public-sector cost remain. Purchases create reserve liabilities; a price-stability regime remunerates those reserves at the policy rate, changing the maturity of the public liability while preserving the consolidated interest expense.

The capacity boundary should therefore trigger explicit alternatives:

1. **Price stability with fiscal closure.** The Federal Reserve maintains the inflation response while Congress changes taxes or primary spending.
2. **Sterilized market support.** The Federal Reserve buys Treasuries for market functioning and continues paying the shadow policy rate on reserves.
3. **Fiscal dominance or a yield cap.** Treasury yields are suppressed while the inflation and credibility consequences enter the macro path.
4. **Forced or subsidized absorption.** Regulation, capital controls, guarantees, or balance-sheet mandates expand demand and reallocate the loss.
5. **Payment disruption or restructuring.** Contractual claims change after the other margins prove unavailable.

The recovery chronology selects the first alternative and uses temporary market support as a bridge. A separate run carries the central private-clearing path through 2039Q2 and starts the conditional fiscal closure in 2039Q3. The assumed fall in marginal yields represents the combined effect of the facility and the enacted package; Federal Reserve purchases and reserve creation are outside that recovery ledger. A future endogenous regime switch can replace those declared post-settlement paths with policy and investor behavior.

## 10 A conditional recovery after 2039

The recovery run inherits every Treasury cohort, its principal, coupon, and maturity from the last successful quarter of the central private-clearing path. It then publishes the economic environment in which the settlement must work:

| Recovery element        | Conditional treatment                                                                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inherited state         | 2039Q2 debt/GDP 178.82 percent; inflation 4.89 percent; interest expense 20.49 percent of GDP; Fed short-rate setting 8.72 percent.                       |
| Real activity           | Real growth of $-1$ percent through 2040Q3, followed by a recovery to 2.5 percent and a gradual convergence to 1.7 percent in 2056.                       |
| Inflation               | Peaks at 5.2 percent in 2040Q3, then declines to 4.5 percent in 2042Q3, 3 percent in 2050Q3, and 2.5 percent in 2056Q3.                                   |
| Marginal Treasury rates | Bills begin at 8.5 percent and long bonds at 9 percent, then decline gradually to 4.3 and 5.1 percent. Inherited coupons remain unchanged until maturity. |
| Fiscal instrument       | Permanent improvement in the primary balance as a share of contemporaneous GDP, phased in linearly over sixteen quarters.                                 |
| Success condition       | 2056Q3 debt/GDP no higher than its 2039Q2 level, with an increase of no more than 0.1 percentage point over the final four quarters.                      |

The solver requires a 5.48-percent-of-GDP permanent primary-balance improvement. Debt and interest continue rising at first because the government has already locked in crisis rates. The stock reaches 209.27 percent of GDP in 2043Q2, three years after inflation peaks. The primary surplus is then 3.87 percent of GDP, new bill rates have fallen to 5.24 percent, and the debt ratio turns downward even though the total budget remains in deficit.

By 2056Q3, the debt ratio has returned to 178.82 percent and is falling 2.55 percentage points per year. Interest expense has declined from its 21.01-percent peak to 8.88 percent of GDP. Inflation is 2.5 percent, and the primary surplus is 4.03 percent of GDP. Applying the same assumed growth, inflation, and rate normalization without the fiscal improvement produces 265.14 percent debt/GDP in 2056. A restored market and lower marginal yields therefore buy time; the durable primary adjustment turns that time into stabilization.

The success condition deliberately determines the fiscal number. Post-settlement growth, inflation, and yields are scenario choices. The probability that Congress can enact the package and investors will reward it with the assumed rates remains unestimated. The result is a conditional existence proof grounded in the actual inherited coupon structure: America can recover from the 2039 state under a package of this scale, while the recovery takes seventeen years and leaves interest expense far above its pre-crisis share.

## 11 How large are the adjustment alternatives?

The conditional recovery and earlier cohort experiments supply scale comparisons for fiscal and inflationary adjustment. Each row carries the relevant inherited Treasury stock while holding its stated policy paths fixed.

| Conditional adjustment                                                           | Model result                                                                                                                                          |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Joint recovery beginning at the 2039 capacity boundary                           | 5.48-percent-of-GDP permanent primary-balance improvement phased in over four years; 2056 debt/GDP 178.82 percent and falling.                        |
| Immediate permanent primary-balance improvement                                  | 5.99 percent of GDP beginning in 2026Q4 under the ten-year 500-basis-point stress.                                                                    |
| All adjustment assigned to a wage base equal to 42 percent of GDP                | Static 14.25-point wage surcharge; approximately \$950 per month on an \$80,000 wage.                                                                 |
| Return new-issuance rates to baseline after the stressed 2036 stock is inherited | 2046 debt/GDP reaches 227.6 percent and remains on an upward path.                                                                                    |
| Five-year inflationary price-level reset after 2036                              | Returns 2046 debt/GDP to 172.1 percent, while debt/GDP rises 3.37 points in the final year.                                                           |
| Ten-year endpoint and trend stabilization after 2036                             | Average GDP-price inflation of 5.38 percent and 38.64 percent more cumulative price growth than baseline, with new-issuance rates held near baseline. |

The wage surcharge is a static incidence illustration. Labor supply, avoidance, wage bargaining, consumption, and investment would change the revenue yield; CBO’s analysis of large labor-tax adjustments shows that behavioral erosion can require rates to rise over time [5]. A broader tax base, spending reductions, faster productivity growth, or a credibility-induced decline in the risk premium would distribute the adjustment differently.

The inflation experiments distinguish a debt-level reset from flow stabilization. A temporary price increase can lower debt/GDP while the primary and interest deficits continue to push the ratio upward. The ten-year result works because nominal Treasury rates are held below the inflation path long enough to reduce the real return on nominal claims. Expectations, wage indexation, private credit, exchange rates, and the Federal Reserve balance sheet determine whether that rate–inflation configuration can be implemented.

## 12 What is anchored, and what remains assumed

The main quantitative conclusions depend on parameters of very different evidentiary quality. Keeping them separate is essential.

| Component           | Empirical or institutional anchor                                    | Additional model choice                                                             |
|---------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Treasury stock      | Security-level principal, coupon, maturity, TIPS, and FRN data       | Representative new tenors and a fixed issuance mix.                                 |
| CBO path            | Primary deficits, GDP, inflation, and Treasury-rate reference paths  | Quarterly interpolation and post-2036 maturity mapping.                             |
| Interest recipients | CBO Federal Reserve holdings and Treasury TIC foreign holdings       | Aggregate residual allocation; holdings stay fixed.                                 |
| Debt–yield effect   | Published 1–2 bp neutral-rate and 2–3 bp term-premium ranges         | Lagged realized debt gap and maturity-specific translation.                         |
| Interest demand     | Redistribution and interest-exposure literature supports the channel | 25-percent domestic and 5-percent foreign spending propensities.                    |
| Macro feedback      | Signs and lags reflect standard monetary and fiscal transmission     | Reduced-form persistence, multipliers, and pass-through coefficients.               |
| Financing capacity  | Reference gross issuance is observed in the model                    | Yield elasticities and 1.25, 1.5, and 2 times hard ceilings.                        |
| Recovery settlement | Inherited 2039 cohorts and all subsequent Treasury cash accounting   | Assumed growth, inflation, marginal yields, and a terminal stabilization condition. |

The model omits endogenous expectations, exchange rates, capital flight, bank and dealer balance sheets, collateral spirals, maturity-specific investor demand, political response, default dynamics, and general equilibrium. Crisis probability remains outside the model. Its 99- percent rate cap is a numerical boundary; the extreme full-spending path reaches it and is reported only as a stress failure.

These omissions cut in different directions. Fiscal action or a credible monetary-fiscal package could reduce the premium and stop the loop. Expectations, currency depreciation, forced selling, or a shrinking tax base could accelerate it. The reported dates should move when the interest-spending and financing-capacity parameters are estimated more carefully.

## 13 Conclusion

The model answers the project’s original question in four steps. Treasury accounting can carry debt forward indefinitely whenever every auction is guaranteed to clear. The zero-spending control demonstrates that property: debt reaches 252 percent of GDP while inflation stays near 2 percent. An accounting ledger therefore cannot discover a national breaking point by itself.

Economic feedback supplies the pressure missing from that ledger. At the central 15.67-percent spendable-interest share, inflation remains above 4 percent beginning in 2037Q3 and reaches 8.04 percent in 2056. Debt-dependent yields intensify the loop: the unchanged-policy path reaches 11.26 percent inflation and 290 percent debt/GDP. Every tested Fed response at the central spending share leaves inflation pressure rising because higher rates eventually enlarge Treasury’s interest transfer enough to work against private-demand restraint.

Finite buyer capacity turns that pressure into an event. The central curve reaches its limit in 2039Q3 when debt-dependent yields are active. Tight and wide curves move the event from 2034Q3 to beyond 2056, exposing the importance of an empirical estimate for Treasury absorption. At the boundary, continued business as usual has lost one of its hidden premises: private portfolios will no longer accept every additional security at some higher yield.

The conditional recovery supplies the final step. Washington bridges the 2039 auction, enacts a permanent 5.48-percent-of-GDP primary-balance improvement over four years, and earns a gradual normalization of marginal rates. Inflation peaks at 5.2 percent and the economy contracts for a year. Debt keeps rising to 209 percent of GDP in 2043 as inherited coupons work through the budget, then falls to 179 percent in 2056 with inflation at 2.5 percent. Interest expense remains 8.9 percent of GDP, so the recovered country still operates under a much tighter and more contentious fiscal compact. Rate normalization without the fiscal package leaves debt at 265 percent of GDP under the same favorable macro assumptions.

America survives this history because the regime changes before double-digit inflation and near-20-percent policy rates arrive. The adjustment redistributes resources on a historic scale, leaves a higher price level and an unprecedented debt stock, and permanently narrows the federal government’s choices. It also preserves functioning auctions, positive real growth, declining inflation, and an improving debt ratio. That relatively happy ending is available in the model, but delay makes it progressively larger, harsher, and less credible.

## Reproducibility

Run:

```
uv run python scripts/run_2026_household_scenario.py
uv run python scripts/run_2036_monetary_backstop_experiment.py
uv run python scripts/run_sequential_crisis_experiment.py
uv run python scripts/run_sequential_policy_frontier.py
uv run python scripts/run_debt_yield_feedback_experiment.py
uv run python scripts/run_private_market_clearing_experiment.py
uv run python scripts/run_recovery_settlement_experiment.py
```

Scenario configurations are stored under `config/scenarios/`; the interest-recipient benchmark is `config/calibration/interest_recipient_shares_2026.json`; generated paths and summaries are under `data/processed/`. The sequential, debt-yield, policy-frontier, and market-capacity outputs carry the corresponding experiment name in their filename. The debt-yield experiment also exports all 120 quarters of the selected central path and its inflation reference to `docs/confidence_shock_central_plot_data.csv`; Figures 1–3 read that file directly. The recovery experiment exports `docs/recovery_settlement_plot_data.csv` for Figure 4.

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